248205: siliciclastic schist, Julimar prospect

(South West Terrane, Yilgarn Craton)

Location and sampling

PERTH (SH 50-14), WOOROLOO (2134)
MGA Zone 50, 424600E 6513271N

Sampled on 30 September 2020

This sample was collected from the 382.50 – 384.24 m depth interval of diamond drillcore JD018, drilled in 2020 by Chalice Mining Ltd at their Julimar PGE–Ni–Cu–Co–Au project (Chalice Mining Ltd, 2022). The drillhole is located about 13.2 km southeast of the southern end of Lake Chittering, 1.8 km south of Julimar Road, and 1.4 km east of Keating Road.

Geological context

The unit sampled is a siliciclastic schist within the South West Terrane of the Yilgarn Craton (Quentin de Gromard et al., 2021). The schist represents metasedimentary rocks in the hanging wall of the Julimar complex, which is being explored for PGE–Ni–Cu–Co–Au mineralization (Chalice Mining Ltd, 2022). The Julimar complex is a 26 km-long layered mafic intrusion, which is almost entirely under cover and coincident with north- to northeast-trending magnetic features. A pegmatitic leucogabbro from the Gonneville intrusion, a component of the Julimar complex, about 0.8 km to the south-southeast of the present sample, yielded an igneous crystallization age of 2668 ± 4 Ma (GSWA 203747, Wingate et al., 2021a). A sample of granodiorite that cross-cuts serpentinized ultramafic rocks of the Gonneville intrusion about 1.1 km to the south-southeast yielded an igneous crystallization age of 2663 ± 8 Ma and contained xenocrystic zircons dated at c. 3276 and 3206 Ma (GSWA 248207, Wingate et al., 2021c). A U–Pb zircon age of 2664 ± 6 Ma for the Coates Siding gabbro, about 20–30 km to the southeast, was reported in a conference abstract (Wilde and Pidgeon, 2006). Re–Os model ages obtained for a sample of pentlandite–pyrrhotite–chalcopyrite from the Gonneville intrusion suggested that alteration or recrystallization of ore minerals was younger than c. 2356 Ma (GSWA 248203, Wingate et al., 2021b). The present sample was collected to characterize further the geological setting of the Julimar complex.

Petrographic description

The sample is a strongly foliated, fine-grained mica–epidote schist (Fig. 1), consisting of about 50–55% biotite, chlorite and white mica, 40–45% quartz, 3% epidote, 1–2% magnetite and titanite, <1% pyrite, and trace zircon. The schistosity is defined by anastomosing phyllosilicate layers up to 1 mm wide, some of which coalesce into wider mica-rich zones and/or wrap around quartz-rich lenses. Biotite forms subhedral laths up to 0.2 mm long, and is partially replaced by pale green, massive to fibrous chlorite. White mica within micaceous layers occurs as well-formed laths about 0.1 mm long, some parallel to the foliation and some at various angles, including crosscutting chlorite and occurring as lenses parallel to the axes of later kinks. The peak metamorphic assemblage has retrogressed from biotite to chlorite and muscovite. Some quartz lenses are composed entirely of recrystallized polycrystalline quartz, although many also contain significant mica (mostly white mica) and retain a granular texture that suggests they represent boudins of originally continuous fine sandy layers within a predominantly pelitic rock. White mica within these layers forms fine-grained, non-aligned sericite masses between quartz grains. Quartz grains are equant, strained, and average 0.1 mm in size, with slightly serrated boundaries with adjacent micas and rare, smoothly curved original detrital boundaries. Polycrystalline quartz lenses include strained and subgrained quartz up to 4 mm long, but are composed mainly of smaller recrystallized grains with finely serrated mutual boundaries. Some polycrystalline lenses may be relics of early, folded quartz veins. Epidote porphyroblasts are subhedral, up to 0.3 mm in size, and wrapped by the micaceous foliation. Narrow lenses of relict magnetite are up to 0.3 mm long, fringed by titanite, and aligned with the foliation. Pyrite is subhedral to euhedral, forms short seams and grains up to 0.4 mm in size, and appears to overprint the foliation. The protolith was probably a thinly bedded sedimentary rock, composed of alternating pelitic and psammitic layers, that has been deformed and metamorphosed under greenschist facies (biotite zone) conditions.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown, anhedral to euhedral, and variably rounded. The crystals are up to 250 µm long, and equant to elongate, with aspect ratios up to 5:1. Some crystals have pitted outer surfaces and, in cathodoluminescence (CL) images, concentric or planar zoning is truncated at grain boundaries, features consistent with sedimentary transport. A CL image of representative zircons is shown in Figure 2.

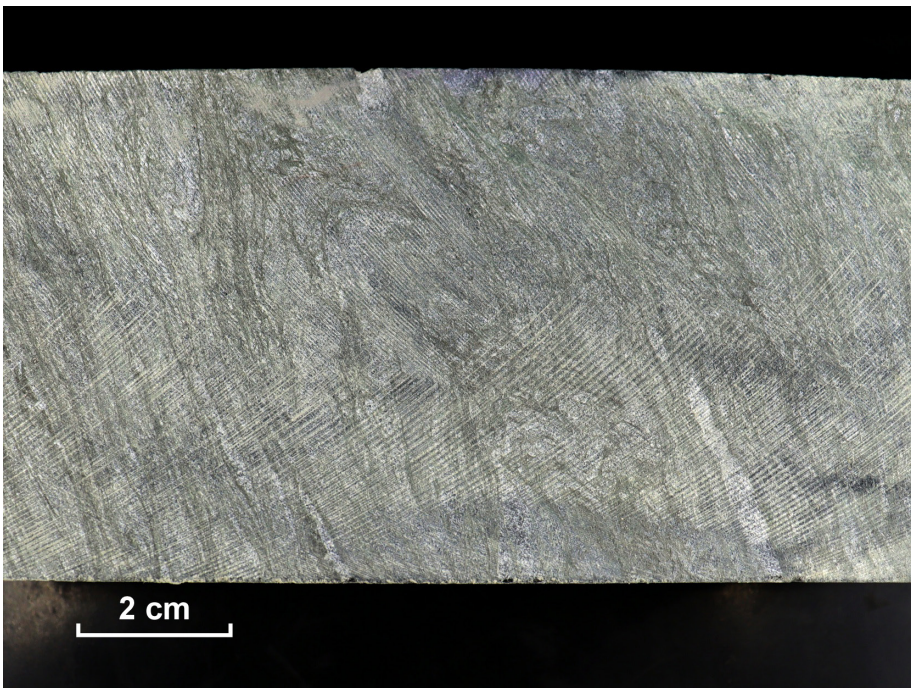

Figure 1. Drillcore image of sample 248205: siliciclastic schist, Julimar prospect

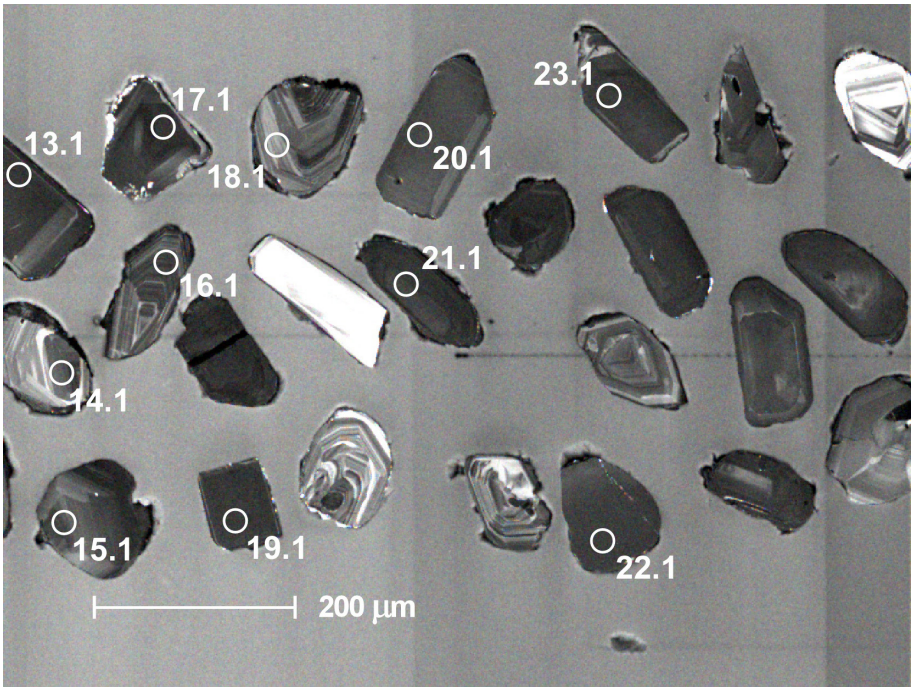

Figure 2. Cathodoluminescence image of representative zircons from sample 248205: siliciclastic schist, Julimar prospect. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 248205: siliciclastic schist, Julimar prospect

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	ƒ204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
Y	59	59.1	610	818	1.34	0.000	2.031	0.018	0.17666	0.00051	2.031	0.018	0.17666	0.00051	2581	20	2622	5	1.6
S	3	3.1	304	250	0.82	0.040	2.006	0.018	0.18026	0.00076	2.007	0.018	0.17990	0.00078	2607	20	2652	7	1.7
S	70	70.1	388	398	1.03	0.023	1.947	0.019	0.18049	0.00066	1.947	0.019	0.18028	0.00067	2672	22	2656	6	-0.6
S	39	39.1	407	429	1.05	-0.013	2.014	0.017	0.18070	0.00069	2.014	0.017	0.18082	0.00070	2599	19	2660	6	2.3
S	63	63.1	190	122	0.64	0.022	1.973	0.022	0.18109	0.00091	1.974	0.022	0.18089	0.00092	2642	24	2661	8	0.7
S	29	29.1	348	306	0.88	0.036	2.002	0.018	0.18122	0.00074	2.002	0.018	0.18090	0.00075	2611	19	2661	7	1.9
S	28	28.1	242	255	1.05	0.011	1.971	0.031	0.18105	0.00088	1.971	0.031	0.18096	0.00088	2645	35	2662	8	0.6
S	58	58.1	362	297	0.82	0.013	1.985	0.019	0.18108	0.00069	1.985	0.019	0.18097	0.00069	2630	21	2662	6	1.2
S	34	34.1	221	226	1.02	0.058	1.979	0.020	0.18160	0.00093	1.980	0.020	0.18108	0.00096	2636	22	2663	9	1.0
S	5	5.1	269	233	0.87	-0.027	1.949	0.018	0.18095	0.00081	1.948	0.018	0.18119	0.00083	2671	21	2664	8	-0.3
S	78	78.1	375	326	0.87	-0.006	2.010	0.019	0.18122	0.00066	2.010	0.019	0.18127	0.00066	2603	21	2665	6	2.3
S	17	17.1	366	418	1.14	0.020	1.973	0.017	0.18151	0.00071	1.973	0.017	0.18133	0.00071	2643	19	2665	7	0.8
S	51	51.1	316	324	1.02	0.063	1.943	0.020	0.18194	0.00077	1.945	0.020	0.18137	0.00079	2675	23	2665	7	-0.4
S	20	20.1	245	244	1.00	-0.010	1.977	0.019	0.18127	0.00087	1.977	0.019	0.18136	0.00087	2639	21	2665	8	1.0
S	8	8.1	369	365	0.99	0.007	1.964	0.017	0.18143	0.00070	1.964	0.017	0.18137	0.00070	2653	19	2666	6	0.5
S	49	49.1	389	376	0.97	0.000	1.940	0.019	0.18141	0.00067	1.940	0.019	0.18141	0.00067	2680	22	2666	6	-0.5
S	54	54.1	178	187	1.05	0.027	1.948	0.023	0.18173	0.00100	1.948	0.023	0.18149	0.00101	2671	26	2667	9	-0.2
S	19	19.1	438	453	1.03	0.023	1.969	0.017	0.18170	0.00066	1.970	0.017	0.18149	0.00067	2647	19	2667	6	0.7
S	41	41.1	340	310	0.91	-0.025	1.955	0.019	0.18129	0.00068	1.954	0.019	0.18151	0.00069	2664	22	2667	6	0.1
S	6	6.1	314	350	1.11	0.000	1.984	0.018	0.18151	0.00074	1.984	0.018	0.18151	0.00074	2631	20	2667	7	1.3
S	77	77.1	506	607	1.20	0.000	1.982	0.018	0.18159	0.00056	1.982	0.018	0.18159	0.00056	2633	20	2667	5	1.3
S	35	35.1	204	120	0.59	0.000	1.995	0.021	0.18167	0.00096	1.995	0.021	0.18167	0.00096	2619	23	2668	9	1.8
S	52	52.1	339	193	0.57	0.030	1.911	0.019	0.18199	0.00130	1.911	0.019	0.18172	0.00131	2713	23	2669	12	-1.7
S	1	1.1	156	87	0.56	0.062	1.955	0.022	0.18229	0.00107	1.957	0.022	0.18174	0.00111	2661	25	2669	10	0.3
S	33	33.1	312	192	0.61	0.041	2.001	0.018	0.18211	0.00078	2.002	0.018	0.18174	0.00080	2612	20	2669	7	2.1
S	76	76.1	489	382	0.78	0.004	1.996	0.019	0.18181	0.00058	1.996	0.019	0.18177	0.00058	2618	20	2669	5	1.9
S	30	30.1	490	516	1.05	-0.010	1.943	0.016	0.18172	0.00062	1.943	0.016	0.18181	0.00063	2677	18	2670	6	-0.3
S	42	42.1	420	430	1.02	0.043	2.005	0.019	0.18234	0.00060	2.006	0.019	0.18195	0.00061	2608	21	2671	6	2.4

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
S	12	12.1	273	278	1.02	0.000	1.983	0.019	0.18202	0.00082	1.983	0.019	0.18202	0.00082	2633	21	2671	7	1.5
S	2	2.1	350	342	0.98	0.007	1.952	0.017	0.18212	0.00072	1.953	0.017	0.18206	0.00072	2666	20	2672	7	0.2
S	25	25.1	284	322	1.14	0.046	1.976	0.019	0.18250	0.00083	1.977	0.019	0.18209	0.00085	2639	21	2672	8	1.3
S	47	47.1	237	271	1.15	0.036	2.003	0.021	0.18246	0.00082	2.003	0.021	0.18214	0.00083	2610	23	2673	8	2.3
S	75	75.1	434	461	1.06	0.005	2.007	0.019	0.18219	0.00059	2.007	0.019	0.18215	0.00059	2606	20	2673	5	2.5
S	32	32.1	400	405	1.01	0.000	1.967	0.017	0.18219	0.00067	1.967	0.017	0.18219	0.00067	2650	19	2673	6	0.9
S	10	10.1	343	292	0.85	0.014	1.975	0.017	0.18234	0.00073	1.975	0.017	0.18222	0.00074	2641	19	2673	7	1.2
S	66	66.1	394	391	0.99	0.005	2.008	0.019	0.18234	0.00064	2.008	0.019	0.18229	0.00064	2605	21	2674	6	2.6
S	27	27.1	349	317	0.91	0.015	2.008	0.018	0.18246	0.00074	2.009	0.018	0.18233	0.00075	2605	19	2674	7	2.6
S	65	65.1	410	407	0.99	0.000	1.975	0.019	0.18236	0.00064	1.975	0.019	0.18236	0.00064	2642	21	2675	6	1.2
S	43	43.1	394	343	0.87	0.011	1.989	0.019	0.18247	0.00065	1.989	0.019	0.18237	0.00065	2625	21	2675	6	1.8
S	23	23.1	299	250	0.83	0.032	1.989	0.018	0.18269	0.00133	1.989	0.018	0.18241	0.00134	2626	20	2675	12	1.8
S	60	60.1	229	127	0.56	0.055	1.937	0.034	0.18304	0.00090	1.938	0.034	0.18255	0.00093	2682	38	2676	8	-0.2
S	79	79.1	424	440	1.04	0.026	1.960	0.019	0.18288	0.00063	1.960	0.019	0.18265	0.00064	2657	21	2677	6	0.7
S	7	7.1	415	297	0.71	0.000	1.979	0.017	0.18274	0.00067	1.979	0.017	0.18274	0.00067	2636	19	2678	6	1.6
S	9	9.1	291	291	1.00	0.000	1.974	0.018	0.18277	0.00080	1.974	0.018	0.18277	0.00080	2642	20	2678	7	1.4
S	13	13.1	267	255	0.95	0.000	1.988	0.019	0.18300	0.00136	1.988	0.019	0.18300	0.00136	2627	21	2680	12	2.0
S	67	67.1	199	191	0.96	0.000	1.929	0.023	0.18305	0.00203	1.929	0.023	0.18305	0.00203	2693	27	2681	18	-0.5
S	14	14.1	205	126	0.62	0.040	1.681	0.017	0.22824	0.00242	1.682	0.017	0.22789	0.00243	3008	25	3037	17	0.9
S	74	74.1	149	115	0.77	0.023	1.677	0.020	0.22818	0.00112	1.678	0.020	0.22797	0.00113	3014	29	3038	8	0.8
S	40	40.1	393	308	0.78	0.035	1.683	0.015	0.24396	0.00080	1.684	0.015	0.24365	0.00080	3005	21	3144	5	4.4
S	46	46.1	296	288	0.98	0.037	1.685	0.017	0.24450	0.00081	1.686	0.017	0.24417	0.00082	3003	24	3147	5	4.6
S	24	24.1	136	103	0.76	0.044	1.625	0.019	0.24494	0.00133	1.626	0.020	0.24455	0.00135	3090	30	3150	9	1.9
S	73	73.1	113	101	0.89	0.000	1.591	0.032	0.24662	0.00135	1.591	0.032	0.24662	0.00135	3144	50	3163	9	0.6
S	22	22.1	325	81	0.25	0.046	1.598	0.014	0.24748	0.00085	1.598	0.014	0.24708	0.00086	3133	23	3166	6	1.1
S	57	57.1	344	322	0.94	0.014	1.619	0.016	0.24721	0.00077	1.620	0.016	0.24709	0.00077	3100	25	3166	5	2.1
S	38	38.1	151	167	1.11	0.013	1.606	0.019	0.24733	0.00126	1.606	0.019	0.24722	0.00127	3120	29	3167	8	1.5
S	15	15.1	194	70	0.36	0.000	1.569	0.017	0.24793	0.00109	1.569	0.017	0.24793	0.00109	3178	28	3172	7	-0.2

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	ƒ204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
S	61	61.1	74	88	1.19	-0.022	1.633	0.024	0.24777	0.00168	1.632	0.024	0.24796	0.00169	3080	36	3172	11	2.9
S	80	80.1	262	185	0.71	0.000	1.561	0.016	0.25767	0.00094	1.561	0.016	0.25767	0.00094	3191	27	3233	6	1.3
S	48	48.1	196	90	0.46	0.027	1.542	0.017	0.26102	0.00111	1.542	0.017	0.26079	0.00112	3222	29	3252	7	0.9
S	36	36.1	205	152	0.74	0.034	1.582	0.016	0.26130	0.00108	1.583	0.016	0.26100	0.00109	3157	26	3253	7	3.0
S	53	53.1	310	228	0.73	0.005	1.548	0.016	0.26133	0.00084	1.548	0.016	0.26129	0.00084	3212	26	3255	5	1.3
S	68	68.1	201	88	0.44	0.044	1.562	0.017	0.26288	0.00101	1.563	0.017	0.26249	0.00102	3188	28	3262	6	2.3
S	45	45.1	226	145	0.64	0.059	1.596	0.017	0.26460	0.00092	1.596	0.017	0.26409	0.00093	3135	26	3271	6	4.2
S	55	55.1	114	89	0.78	0.062	1.565	0.020	0.26550	0.00131	1.566	0.020	0.26496	0.00134	3183	32	3277	8	2.9
S	72	72.1	169	93	0.55	0.034	1.552	0.017	0.26565	0.00179	1.552	0.017	0.26536	0.00180	3205	29	3279	11	2.2
S	50	50.1	333	182	0.55	0.024	1.472	0.015	0.26562	0.00084	1.473	0.015	0.26540	0.00085	3341	27	3279	5	-1.9
S	4	4.1	82	28	0.35	0.081	1.558	0.022	0.26733	0.00168	1.559	0.022	0.26663	0.00172	3195	36	3286	10	2.8
S	11	11.1	100	70	0.70	-0.051	1.476	0.020	0.27036	0.00156	1.475	0.020	0.27080	0.00158	3336	35	3311	9	-0.8
S	26	26.1	198	133	0.67	-0.017	1.524	0.016	0.27111	0.00114	1.524	0.016	0.27126	0.00114	3252	27	3313	7	1.9
S	64	64.1	249	207	0.83	-0.006	1.482	0.016	0.27189	0.00097	1.482	0.016	0.27194	0.00097	3324	28	3317	6	-0.2
S	44	44.1	96	65	0.68	0.045	1.580	0.021	0.27254	0.00149	1.580	0.021	0.27215	0.00151	3160	34	3318	9	4.8
S	21	21.1	315	43	0.14	-0.027	1.515	0.014	0.27255	0.00090	1.514	0.014	0.27279	0.00091	3269	24	3322	5	1.6
S	69	69.1	155	53	0.34	0.010	1.512	0.018	0.27442	0.00122	1.512	0.018	0.27433	0.00122	3272	31	3331	7	1.8
S	16	16.1	302	147	0.49	0.023	1.484	0.014	0.27519	0.00094	1.484	0.014	0.27499	0.00094	3321	24	3335	5	0.4
S	18	18.1	100	37	0.37	-0.034	1.507	0.020	0.27728	0.00161	1.507	0.020	0.27758	0.00162	3282	34	3349	9	2.0
S	71	71.1	88	41	0.47	0.030	1.497	0.020	0.27902	0.00153	1.497	0.020	0.27876	0.00154	3298	36	3356	9	1.7
S	62	62.1	153	69	0.45	0.063	1.469	0.017	0.27972	0.00119	1.470	0.017	0.27918	0.00121	3345	31	3358	7	0.4
S	37	37.1	91	31	0.34	0.019	1.459	0.021	0.28563	0.00174	1.459	0.021	0.28547	0.00175	3365	38	3393	10	0.8
S	31	31.1	183	105	0.57	0.060	1.330	0.014	0.34434	0.00131	1.331	0.014	0.34385	0.00132	3613	30	3680	6	1.8
D	56	56.1	528	701	1.33	0.017	2.263	0.021	0.17050	0.00053	2.264	0.021	0.17035	0.00054	2359	18	2561	5	7.9

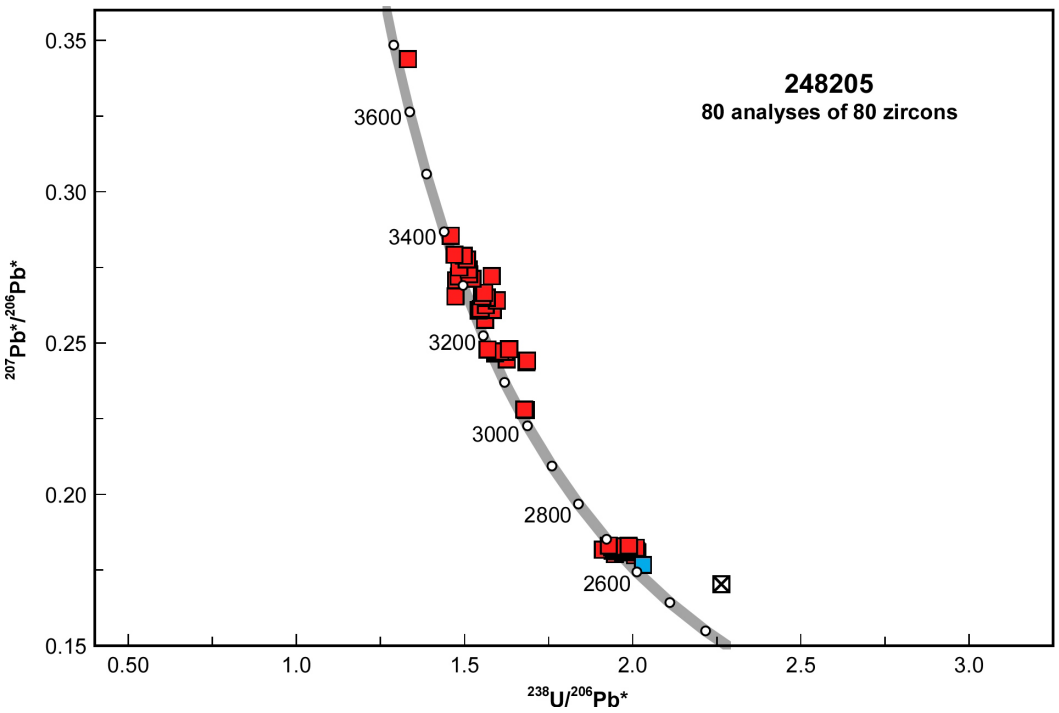

Figure 3. **U–Pb analytical data for sample 248205: siliciclastic schist, Julimar prospect. Blue square indicates Group Y (youngest detrital zircon); red squares indicate Group S (older detrital zircons); crossed square indicates Group D (discordance >5%)**

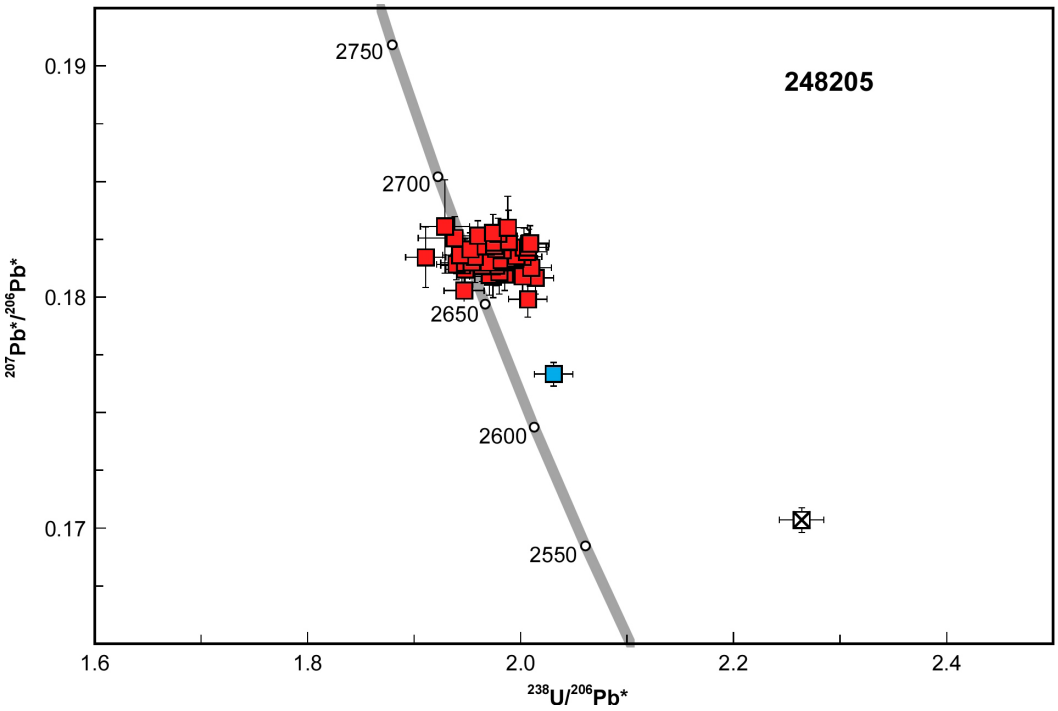

Figure 4. **Expanded view of U–Pb analytical data for sample 248205: siliciclastic schist, Julimar prospect. Symbols as in Figure 3**

Analytical details

This sample was analysed on 19–20 and 25–26 November 2021, using SHRIMP-B. Analyses 1.1 to 40.1 (spot numbers 1–40) were obtained during the first session, together with 10 analyses of the M257 standard, which indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.21% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the first session was corrected by reference to the OGC1 standard; measured ratios were decreased by 0.04%. Analyses 41.1 to 80.1 (spot numbers 41–80) were obtained during the second session, together with 17 analyses of the M257 standard, of which 14 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.70% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.25% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Eighty analyses were obtained from 80 zircons. Results are listed in Table 1, and shown in concordia diagrams (Figs 3 and 4) and a probability density plot (Fig. 5).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). One analysis is >5% discordant. The date obtained from this analysis (Group D; Table 1) is unreliable, and considered not to be geologically significant. The remaining 79 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group Y comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 2622 ± 5 Ma (1σ).

Group S comprises 78 analyses (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 3680–2652 Ma.

It is possible that all of the analyses are of unmodified detrital zircons, in which case the date of 2622 ± 5 Ma (1σ) for the single analysis in Group Y represents a maximum depositional age for the sandstone. However, the high U and Th content of this zircon suggests the date could instead reflect loss of radiogenic Pb. A more conservative estimate of the maximum depositional age can be based on the weighted mean date of 2669 ± 2 Ma (MSWD = 0.75) for the youngest 45 analyses in Group S.

The data for combined Groups Y and S indicate significant age components at c. 3356, 3318, 3276, 3254, 3166, 3146, and 2669 Ma, based on contributions from approximately 3, 5, 5, 4, 6, 3, and 43 analyses, respectively (Fig. 5). These are interpreted as the ages of zircon-crystallizing rocks in the detrital source region(s), or as the ages of detrital components within sediments that have been reworked into this rock.

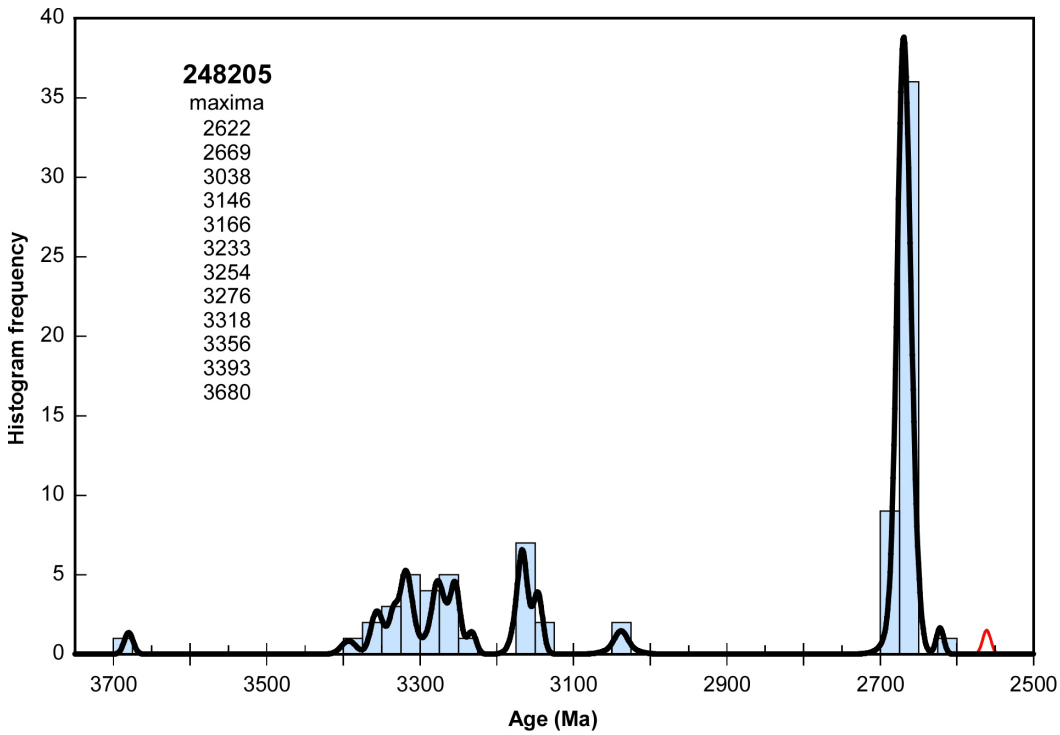

Figure 5. Probability density diagram and histogram for sample 248205: siliciclastic schist, Julimar prospect. Thick curve, maxima values, and blue histogram (bin width 25 Ma) include only data with discordance <5% (79 analyses of 79 zircons). Thin curve includes all data (80 analyses of 80 zircons)

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Recommended reference for this publication

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Data obtained: 26 November 2021

Data released: 10 October 2022